Analytical Generalized Impulse Responses and Connectedness in Score-Driven Models*

Job market paper

Emilija Dzuverovic Dario Palumbo

Ca' Foscari University of Venice

*Dario Palumbo developed the score-driven MGB2 model and proposed extending the connectedness framework to this setting. Emilija Dzuverovic derived the analytical generalized impulse responses and their implications for shock transmission and connectedness, and carried out the simulations and empirical applications.

Motivation

Connectedness measures summarize how strongly markets respond to shocks arising in other markets (Diebold and Yilmaz, 2009, 2012, 2014).

They measure relative contributions, rather than absolute response magnitudes.

Magnitudes

Similar connectedness values can be associated with substantially different response magnitudes.

Timing

Cumulative measures can obscure whether responses occur on impact or persist at later horizons.

How can we make the link between connectedness measures and the magnitude and timing of shock responses explicit?

This paper

Analytical generalized impulse responses for a nonlinear score-driven model of strictly positive variables, such as realized volatility.

Main result:

$$\textbf{Generalized response} = \textbf{baseline forecast} \times \textbf{relative shock effect}$$

Market conditions affect the baseline forecast. The relative effect of the shock is independent of the filtered history.

- **Transmission:** conditions under which cross-series responses vanish and how direct and indirect effects arise within this model.
- **Connectedness:** how market conditions affect the measures and why impact and later horizons should be examined separately.

Evidence: simulations; EUR/USD and GBP/USD (2012-2026); four futures markets.

Related literature

- **Generalized impulse response function (GIRF)**
  Koop, Pesaran and Potter (1996); Pesaran and Shin (1998).
- **Generalized forecast error variance decomposition (GFEVD)**
  Lanne and Nyberg (2016).
- **Connectedness measures**
  Diebold and Yilmaz (2009, 2012, 2014).
- **Score-driven models**
  Creal, Koopman and Lucas (2013); Harvey (2013); Harvey and Palumbo (2023); Palumbo (2026).

Our contribution: analytical responses for a nonlinear model based on the flexible multivariate generalized beta of the second kind (MGB2) distribution, with explicit implications for transmission and connectedness.

Advantages of the framework

- **Compared with simulation-based nonlinear GIRFs:**
  Analytical responses avoid simulating future observation paths.
- **Compared with a Gaussian VAR in logarithms:**
  We model strictly positive variables directly through nonlinear score-driven dynamics based on the flexible MGB2 distribution.
- **Interpretation:**
  The analytical formulas make explicit how model dynamics and market conditions affect responses and connectedness.

Distributional flexibility with analytical tractability.

Model and analytical responses

Score-driven MGB2 model

Following Palumbo (2026), for strictly positive $y_t = (y_{1t}, \ldots, y_{Nt})'$:

$$y_t = D_t \varepsilon_t, \qquad D_t = \mathrm{diag}(\alpha_{1,t|t-1}, \ldots, \alpha_{N,t|t-1}), \qquad \alpha_{i,t|t-1} = \exp(\lambda_{i,t|t-1}).$$

The innovation vectors ε_t are i.i.d. MGB2 with shape parameters

$$\xi = (\xi_1, \ldots, \xi_N)', \qquad \nu = (\nu_1, \ldots, \nu_N)', \qquad \varsigma > 0.$$

The vector of log-scales follows

$$\lambda_{t+1|t} = (I - \Phi_\lambda)\omega_\lambda + \Phi_\lambda \lambda_{t|t-1} + K_\lambda u_t,$$

$$u_t = D_\nu (c\, b_t - \xi), \qquad D_\nu = \mathrm{diag}(\nu_1, \ldots, \nu_N), \qquad c = \varsigma + \sum_{i=1}^{N} \xi_i,$$

with

$$b_{it} = \frac{\varepsilon_{it}^{\nu_i}}{1 + \sum_{j=1}^{N} \varepsilon_{jt}^{\nu_j}}, \qquad b_{0t} = \frac{1}{1 + \sum_{j=1}^{N} \varepsilon_{jt}^{\nu_j}}.$$

- $(b_{1t}, \ldots, b_{Nt}, b_{0t}) \sim \mathrm{Dirichlet}(\xi_1, \ldots, \xi_N, \varsigma)$: bounded score with $\mathbb{E}(u_t \mid \mathcal{F}_{t-1}) = 0$.

Generalized shocks and responses

A shock to series i conditions on

$$z_{it} = \varepsilon_{it} - \mu_i = \delta_i, \qquad \mu_i = \mathbb{E}(\varepsilon_{it}), \qquad \mu_i + \delta_i > 0.$$

The generalized impulse response is

$$\mathrm{GIRF}_i(h; \delta_i, \mathcal{F}_{t-1}) = \mathbb{E}\big[y_{t+h} \mid \mathcal{F}_{t-1}, z_{it} = \delta_i\big] - \mathbb{E}\big[y_{t+h} \mid \mathcal{F}_{t-1}\big], \qquad h = 0, \ldots, H.$$

Koop, Pesaran and Potter (1996); Pesaran and Shin (1998)

Order-invariant

Responses do not depend on the ordering of the series.

Preserves dependence

Other innovations are integrated out under their conditional MGB2 distribution.

Reduced form

The shock is not an independently identified structural intervention.

Analytical GIRFs

Let $m_{j,t}(h) = \mathbb{E}[y_{j,t+h} \mid \mathcal{F}_{t-1}]$ be the baseline forecast. For every $h \geq 0$,

Proposition 1

$$\mathrm{GIRF}_{j \leftarrow i}(h; \delta_i, \mathcal{F}_{t-1}) = \underbrace{m_{j,t}(h)}_{\text{baseline forecast}} \times \underbrace{g_{ji}(h; \delta_i)}_{\text{relative response}},$$

where g_{ji} does not depend on $\mathcal{F}_{t-1}$ and is given by

$$g_{ii}(0; \delta_i) = \delta_i / \mu_i, \qquad g_{ji}(0; \delta_i) = \mathbb{E}\big(\varepsilon_{jt} \mid \varepsilon_{it} = \mu_i + \delta_i\big)/\mu_j - 1, \quad j \neq i,$$

$$g_{ji}(h; \delta_i) = \frac{M_u^{(i)}\big(s_j(h-1); q_i\big)}{M_u\big(s_j(h-1)\big)} - 1, \qquad h \geq 1, \qquad s_j(r) = K_\lambda' (\Phi_\lambda')^r e_j.$$

The bounded score ensures that the moment generating functions (MGFs) are finite. We evaluate them numerically, without simulating future observation paths.

Analytical responses are checked against independent simulations. See Appendix C of the paper.

When do cross-series responses vanish?

Corollary 1

Let $h \geq 1$ and $j \neq i$. If $(\Phi_\lambda^{h-1} K_\lambda)_{ji} = 0$, then $\mathrm{GIRF}_{j \leftarrow i}(h; \delta_i, \mathcal{F}_{t-1}) = 0$ for every admissible shock size and every filtered history.

With diagonal Φ_λ, the score u_{it} enters the response index of series j with weight $\phi_j^{h-1} k_{ji}$:

- $k_{ji} \neq 0$ can generate a direct response that persists through series j's own dynamics.
- If $k_{31} = 0$, $k_{21} \neq 0$ and $k_{32} \neq 0$ do not generate a response of series 3 to a shock in series 1 at $h \geq 1$.
- Off-diagonal elements of Φ_λ can allow a shock to affect another series through an intermediate series.

Corollary 3

If Φ_λ and K_λ are diagonal, spillovers are purely contemporaneous.

Connectedness from the analytical responses

Let us draw $\varepsilon^{(b)}$ from the fitted joint MGB2 distribution and set $\delta^{(b)} = \varepsilon^{(b)} - \mu$. For a shock vector δ and horizon H, (Lanne and Nyberg, 2016):

$$\lambda_{ji,t}^{(H)}(\delta) = \frac{\sum_{h=0}^{H} m_{j,t}(h)^2 \, g_{ji}(h;\delta_i)^2}{\sum_{r=1}^{N} \sum_{h=0}^{H} m_{j,t}(h)^2 \, g_{jr}(h;\delta_r)^2}, \qquad \theta_{ji,t}^{(H)} = \mathbb{E}_{\delta}\big[\lambda_{ji,t}^{(H)}(\delta) \mid \mathcal{F}_{t-1}\big].$$

- Filtered history enters the GFEVD only through $m_{j,t}(h)/m_{j,t}(0)$.
- Total, directional and net measures as in Diebold and Yilmaz (2014).
- $B = 32{,}000$ shock draws, with separate normalization for impact $h = 0$, after impact $h = 1, \ldots, H$, and cumulative $h = 0, \ldots, H$.

Corollary 2

For $H = 0$, $\lambda_{ji,t}^{(0)}(\delta) = g_{ji}(0;\delta_i)^2 / \sum_r g_{jr}(0;\delta_r)^2$: impact connectedness is constant over time with fixed shape parameters.

Three implications for connectedness

1

Impact connectedness

With fixed MGB2 shape parameters, impact connectedness is constant over time and reflects contemporaneous dependence between innovations.

2

Shock transmission

Score coefficients can generate direct responses. Transmission through intermediate series requires off-diagonal elements of Φ_λ.

3

Market conditions

Market conditions affect response magnitudes, while relative responses remain unchanged. Similar connectedness measures can therefore coexist with substantial differences in response magnitudes.

Report response magnitudes alongside connectedness measures and distinguish responses on impact from those at later horizons.

Simulations

Simulation design

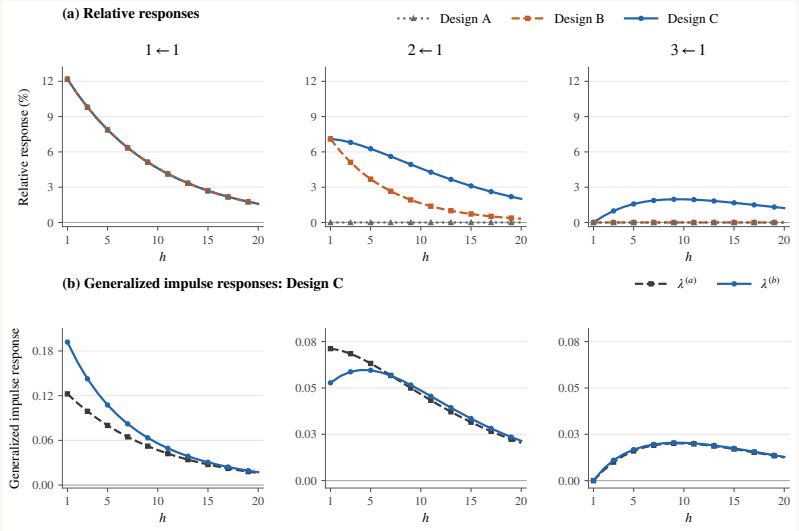
$$\xi = (1.5, 2.0, 1.2)', \qquad \nu = (1.8, 1.3, 2.2)', \qquad \omega_{\lambda,j} = -\log \mu_j.$$

Design	Φ_λ	K_λ	ς
A	Φ_0	K_0	3
B	Φ_0	$K_0 + 0.03L$	3
C	$\Phi_0 + 0.08L$	$K_0 + 0.03L$	3
C ($\varsigma = 2$)	$\Phi_0 + 0.08L$	$K_0 + 0.03L$	2
C ($\varsigma = 6$)	$\Phi_0 + 0.08L$	$K_0 + 0.03L$	6

$\Phi_0 = \mathrm{diag}(0.90, 0.85, 0.80)$, $K_0 = \mathrm{diag}(0.05, 0.04, 0.06)$. $L_{21} = L_{32} = 1$; all other entries are zero.

- Shocks at the 10th and 90th innovation percentiles.
- Initial log-scale vectors: $\lambda^{(a)} = \omega_\lambda$ and $\lambda^{(b)} = \omega_\lambda + (0.5, -0.4, 0.3)'$.
- $B = 32{,}000$ shock draws, common to both histories.

Responses to a 90th-percentile shock to series 1



Top: relative responses in A–C. Bottom: generalized responses in C at two histories.

Cumulative tables can obscure transmission channels

1.5 pp

Difference in total connectedness between Designs A and C ($0 \leq h \leq 20$).

0–32.4%

Range of total connectedness over $1 \leq h \leq 20$.

45.6%

Impact connectedness at $\varsigma = 2$, up from 24.7% at $\varsigma = 6$.

- Impact connectedness is identical in A–C (37.6%) and history independent.
- Excluding impact responses, Design B has cross-series GFEVD shares only for $2 \leftarrow 1$ (48.0%) and $3 \leftarrow 2$ (23.6%). In Design C, shocks to series 1 also affect series 3, with a share of 11.1%.

Exchange rates

EUR/USD and GBP/USD realized variances

Data

5-min realized variances (VOLARE)

$\boldsymbol{T = 3825}$ daily observations

	Mean	s.d.
EUR/USD	0.615	0.663
GBP/USD	0.748	1.531

Model selection

Multivariate Burr ($\xi_i = 1, \forall i$) under three dynamic specifications.

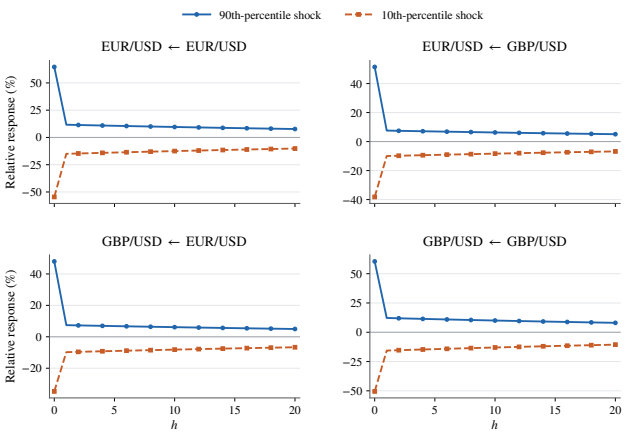
Baseline estimates

	EUR/USD	GBP/USD
ϕ_{jj}	0.979 (0.004)	0.979 (0.004)
k_{jj}	0.049 (0.005)	0.047 (0.006)
k_{12}	0.030 (0.004)	
k_{21}		0.033 (0.004)
$\log \varsigma$	-0.593 (0.036)	
LLF	1,433.35	
BIC	$-2{,}776.0$	

Robust Newey–West standard errors in parentheses.

BIC selects: diagonal Φ_λ and unrestricted K_λ. LLF increases by $\mathbf{209.5}$ relative to diagonal dynamics.

Relative responses



90th-percentile shocks

Responses as percentages of the baseline forecast.

On impact

Shock	EUR/USD	GBP/USD
EUR/USD	65%	48%
GBP/USD	51%	60%

At later horizons

	Own	Cross-series
$h = 1$	12%	7–8%
$h = 20$	8%	5%

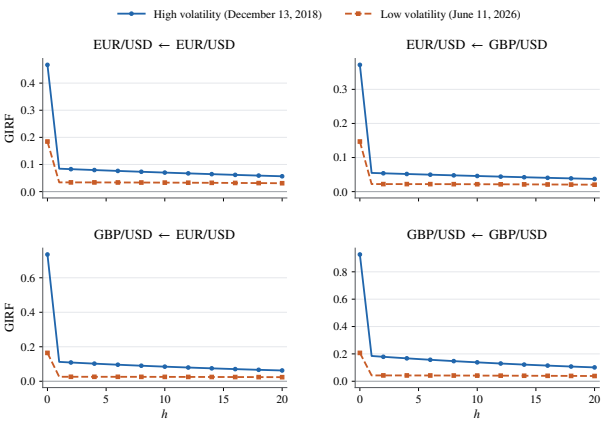
Connectedness

Horizon	EUR ← GBP	GBP ← EUR	Total	Net EUR (pp)
$h = 0$	44.24	43.74	**43.99**	−0.49
$h = 0, \ldots, 10$	40.70	39.38	**40.04**	−1.32
$h = 0, \ldots, 20$	39.96	38.50	39.23	−1.45
$h = 1, \ldots, 10$	36.83	34.84	**35.84**	−1.99
$h = 1, \ldots, 20$	36.90	34.91	35.91	−1.98

- Over time, the total over $h = 0, \ldots, 10$ ranges only from 39.53% to 40.57%.
- The directional imbalance is modest and depends on the specification and horizon.

Shares and total connectedness in %; net EUR in percentage points.
$B = 32{,}000$ draws; MC standard errors at most 0.31 pp.

Similar connectedness, different response magnitudes



Responses to a shock at the same innovation quantile on a high-volatility day (Dec $13, 2018$) vs. a low-volatility day (Jun $11, 2026$):

- EUR/USD: $2.5\times$ on impact, $1.8\times$ at $h = 20$
- GBP/USD: $4.5\times$ on impact, $2.6\times$ at $h = 20$

Gaussian VAR benchmark

VAR(10) in log realized variances, same sample (Diebold and Yilmaz, 2012, 2014)

Model	Measure	Total (%)	Net EUR (pp)
VAR, log RV	Conventional GFEVD, 11 steps	32.90	$+0.02$
VAR, log RV	GIRFs in RV levels, LN shares	39.15	$+0.36$
Burr	GIRFs in RV levels, LN shares	40.04	-1.32

Sample averages. LN shares over $h = 0, \ldots, 10$. VAR responses in RV levels assume Gaussian innovations.

- The difference in total connectedness narrows when the response scale and GFEVD definition are aligned.
- Net connectedness estimates remain sensitive to the model.
- The shocks remain model specific.

Futures markets

Four futures markets

Data

- S&P 500 E-mini (ES), crude oil (CL), natural gas (NG), and gold (GC)
- 5-min realized variances (VOLARE)
- $T = 4242$ daily observations
- Log correlations from 0.26 (NG, GC) to 0.57 (ES, CL)

Estimation

- Multivariate Burr ($\xi_i = 1, \forall i$)

Baseline estimates

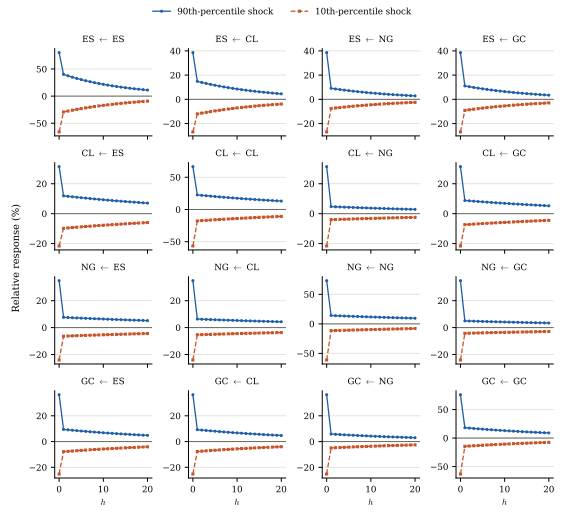
	ES	CL	NG	GC
ϕ_{ii}	0.939	0.973	0.980	0.965
k_{ii}	0.159	0.076	0.055	0.072
$\nu_i\varsigma$	3.44	4.25	3.82	3.66

$\log \varsigma = 0.371$

Heavy upper tails: $\nu_i\varsigma$ between 3.44 and 4.25.

BIC selects: diagonal Φ_λ and unrestricted K_λ. LLF increases by $\mathbf{388.2}$ relative to diagonal dynamics.

Relative responses across futures markets



Responses as percentages of baseline forecasts. Rows: receivers. Columns: shock sources.

Relative responses across futures markets

Responses to 90**th-percentile shocks**, as percentages of the baseline forecast.

Cross-series impact responses

Receiver	Response
ES	38.6%
CL	31.4%
NG	34.9%
GC	36.4%

For each receiver, the impact response is the same across the other sources at a common innovation percentile. This follows from the Burr specification.

Own impact responses

From 67% for CL to 80% for ES.

After impact

- Responses to NG shocks are the smallest.
- Responses to ES shocks are the most persistent.

For example, the response of CL to an ES shock is

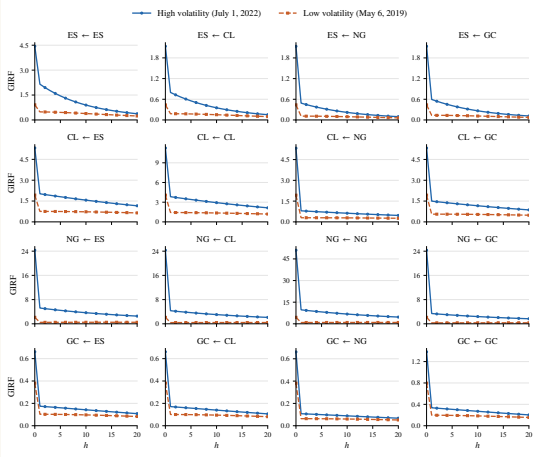
$$11.9\% \text{ at } h = 1, \qquad 7.1\% \text{ at } h = 20.$$

Connectedness across futures markets

Net connectedness (pp)	ES	CL	NG	GC	Total (%)
Burr, $h = 1, \ldots, 10$	$+29.3$	$+9.8$	-26.4	-12.7	45.4
Burr, $h = 0$	-0.6	$+1.6$	-0.4	-0.6	51.7
Burr, $h = 0, \ldots, 10$	$+13.4$	$+4.9$	-12.4	-5.9	48.9
VAR(5), LN shares in RV levels, $h = 0, \ldots, 10$	$+14.4$	$+4.6$	-16.6	-2.3	44.7
VAR(5), conventional GFEVD, 11 steps	$+12.2$	$+0.7$	-9.1	-3.9	29.6

- Beyond impact, ES is the largest net transmitter, followed by CL, whereas NG and GC are net receivers.
- The benchmark model identifies the same net transmitters and receivers; magnitudes differ.

Responses on high- and low-volatility days



90th-percentile shocks. High volatility: July 1, 2022. Low volatility: May 6, 2019.

Baseline forecasts on impact are between **1.7** and **11.4** times as large on the high-volatility day.

Limitations and extensions

Limitations and extensions

Limitations

The Burr distribution used in applications simplifies estimation but limits flexibility in marginal shapes, contemporaneous dependence, and cross-series impact responses.

Extensions

Time-varying shape parameters to capture changes in tail behaviour, contemporaneous dependence, and relative responses.

Accounting for parameter uncertainty in connectedness measures.

Conclusion

- In the score-driven MGB2 model, the GIRF is the product of a baseline forecast (market conditions) and a history-independent relative response (the effect of the shock).
- The framework combines a flexible distribution for positive variables with analytical responses, making explicit how model dynamics and market conditions affect connectedness.
- Simulations: including impact responses can obscure differences in transmission at later horizons.
- Exchange rates: connectedness nearly constant, while responses vary substantially across market conditions.
- Futures: ES and CL net transmitters, NG and GC net receivers over the ten trading days after the shock.

Interpret connectedness alongside response magnitudes and horizons.

References

Creal, D., Koopman, S. J., and Lucas, A. (2013). Generalized autoregressive score models with applications. *Journal of Applied Econometrics*, 28(5), 777–795.

Diebold, F. X., and Yilmaz, K. (2009). Measuring financial asset return and volatility spillovers, with application to global equity markets. *Economic Journal*, 119(534), 158–171.

Diebold, F. X., and Yilmaz, K. (2012). Better to give than to receive: Predictive directional measurement of volatility spillovers. *International Journal of Forecasting*, 28(1), 57–66.

Diebold, F. X., and Yilmaz, K. (2014). On the network topology of variance decompositions: Measuring the connectedness of financial firms. *Journal of Econometrics*, 182(1), 119–134.

Harvey, A. C. (2013). *Dynamic Models for Volatility and Heavy Tails: With Applications to Financial and Economic Time Series*. Cambridge University Press.

Harvey, A. C., and Palumbo, D. (2023). Score-driven models for realized volatility. *Journal of Econometrics*, 237(2).

References

Koop, G., Pesaran, M. H., and Potter, S. M. (1996). Impulse response analysis in nonlinear multivariate models. *Journal of Econometrics*, 74(1), 119–147.

Lanne, M., and Nyberg, H. (2016). Generalized forecast error variance decomposition for linear and nonlinear multivariate models. *Oxford Bulletin of Economics and Statistics*, 78(4), 595–603.

Palumbo, D. (2026). Multivariate score-driven models for strictly positive variables. Unpublished manuscript, Ca' Foscari University of Venice.

Pesaran, M. H., and Shin, Y. (1998). Generalized impulse response analysis in linear multivariate models. *Economics Letters*, 58(1), 17–29.